

Shifaa: An Integrated AI-Powered Clinical Decision Support and Real-Time Patient Monitoring Platform

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Abstract—Modern inpatient clinical environments impose two persistent burdens on hospital staff: the slow, error-prone retrieval of patient information from fragmented Electronic Health Records (EHRs) and the difficulty of continuously monitoring patients for life-threatening events. This paper presents Shifaa, an integrated, privacy-preserving clinical decision support platform that addresses both challenges within a single role-aware system. The platform unifies four AI-powered modules: (1) a four-stage conversational EHR retrieval pipeline in which intent classification and named entity recognition (NER) are powered by Qwen 0.5, a lightweight 0.5-billion-parameter model optimized for structured JSON output, while grounded answer generation employs MedGemma 1.5; (2) a two-stage CNN–BiLSTM–Attention cascade for 12-lead ECG arrhythmia detection; (3) a four-stage computer-vision fall detection pipeline using CTR–GCN; and (4) a hybrid C++/ONNX–Python seizure detection pipeline fusing a VSViG temporal branch with a Cross-Joint Attention classifier via a novel series gate. The conversational assistant achieves 94.1% intent accuracy and 94.0% NER macro-F1 with a hallucination rate below 1%. The ECG cascade reaches 96.10% and 94.92% accuracy on binary and sub-type stages. The fall detector achieves 94.87% F1 with 98.83% recall. The seizure detector achieves AUROC 0.9669 at 37.2 FPS with a mean pre-clinical alert latency of −3.94 s. The entire stack operates locally with no cloud dependency.

Index Terms—Clinical decision support, large language models, retrieval-augmented generation, ECG arrhythmia detection, fall detection, epileptic seizure detection, graph convolutional networks, video transformers, microservices, explainable AI.

I. INTRODUCTION

Modern hospitals generate a continuous and growing stream of patient data spanning structured clinical records, physiological time-series signals, and bedside video. Two persistent challenges arise from this abundance. First, the information a clinician needs is rarely consolidated in one place: answering a routine bedside question such as “Is she allergic to penicillin?” may require navigating several screens of an EHR [1]. Second, the events that most threaten patient safety—cardiac arrhythmias, falls, and epileptic seizures—occur in real time and are easily missed when ward staff are stretched thin. Falls cause approximately 684,000 deaths worldwide each year [2], and between 700,000 and one million in-hospital falls occur annually [3]. Epilepsy affects roughly 50 million people globally [4], and cardiovascular disease remains the leading cause of death worldwide at an estimated 17.9 million deaths per year [5].

This paper presents **Shifaa**, an integrated, privacy-preserving clinical decision support and patient monitoring platform. The primary contributions are:

- 1) A structured conversational pipeline employing Qwen 0.5 for intent classification and NER—enabling reliable JSON-structured output on constrained hardware—and MedGemma 1.5 for grounded answer generation, achieving 94.1% intent accuracy with a hallucination rate below 1% and no task-specific training.
- 2) A two-stage CNN–BiLSTM–Attention ECG arrhythmia cascade reaching 96.10% binary screening accuracy and 94.92% sub-type accuracy on a merged multi-dataset corpus with strict patient-level isolation.

- 3) A four-stage real-time fall detection pipeline integrating YOLOv8 tracking, MobileNetV3 role classification, MediaPipe pose estimation, and CTR–GCN, achieving 94.87% fall F1 with 98.83% recall.

- 4) A hybrid C++/ONNX–Python seizure detection pipeline combining VSViG with Cross-Joint Attention via a novel series gate, achieving AUROC 0.9669 and pre-clinical alerting at −3.94 s mean latency at 37.2 FPS.

- 5) A role-aware microservices platform with a unified alert lifecycle, room-centered nurse scoping, and fully local on-premises deployment including the LLM backend.

II. RELATED WORK

A. Clinical NLP and Medical Question Answering

The dominant paradigms for clinical question answering are fine-tuned domain models and retrieval-augmented generation (RAG). Fine-tuned BioBERT-based extractors [7] require labeled training data and must be retrained as the schema evolves. RAG systems [8] can still hallucinate from retrieved context. MedPaLM 2 [9] achieved strong performance on clinical benchmarks using large LLMs. The present work takes a third path: structured, schema-aware retrieval feeding a strictly grounded medical LLM, requiring no task-specific training while producing fully auditable, citation-backed answers. Using the compact Qwen 0.5 for structured extraction and reserving MedGemma 1.5 for generation enables a practical two-model design for resource-constrained clinical deployments.

B. ECG Arrhythmia Detection with Deep Learning

Hannun et al. demonstrated cardiologist-level arrhythmia classification using a deep residual CNN [10]. Subsequent work on PTB-XL showed that CNN-recurrent architectures generalize well across diverse patient populations [11]. Ribeiro et al. trained a deep neural network on over 2 million ECGs from the CODE-15 corpus [12]. The present work extends these approaches through a two-stage cascade that concentrates the harder multi-class decision on the minority of abnormal recordings, improving both accuracy and efficiency.

C. Fall Detection in Clinical Environments

Skeleton-based action recognition with Graph Convolutional Networks (GCNs) became the dominant paradigm following ST-GCN [13] on the NTU RGB+D benchmark. CTR-GCN [14] improves on earlier GCNs by learning an adaptive joint-relationship topology per feature channel. A key challenge in hospital deployment is distinguishing patient falls from medical staff motion, which few prior systems explicitly model. The present work addresses this through an intermediate role classification stage using MobileNetV3 [15].

D. Video-Based Epileptic Seizure Detection

Xu et al. introduced VSViG [16], a lightweight skeleton-based spatiotemporal Vision GNN demonstrating pre-clinical alerting at a mean latency to clinical onset of -13.1 s. As a standalone segment-level classifier, however, VSViG achieves only AUROC 0.810 on our held-out test split—its accumulative-probability alert mechanism produces wide temporal alert regions with limited window-level discrimination, motivating the need for a more discriminative branch. Zamzam et al. [17] proposed Cross-Joint Attention exploiting frozen ViViT features from Kinetics-400, reporting AUROC 0.921 ± 0.012 (frozen ViViT variant). Despite its stronger segment-level discrimination, the CJ branch operates at 5-second segment resolution and, as our ablation confirms (Table VII), all classification-only head architectures tested produced positive mean latency to clinical onset—meaning alerts fire after clinical onset rather

than before it. The present work fuses both through a novel series gate architecture that achieves pre-clinical alerting that neither branch can attain independently.

III. SYSTEM ARCHITECTURE

A. High-Level Design

Fig. 1 illustrates the complete Shifaa system architecture. The platform follows a four-tier design: a React 18 single-page application (presentation tier), a FastAPI API Gateway (application tier), a PostgreSQL relational database (data tier), and four independent AI inference microservices (AI tier). The gateway handles authentication and authorization, rate limiting, request routing, and logging for all user roles—Doctors, Nurses, Administrators, and Family (view-only). External devices (cameras, ECG sensors) feed directly into the monitoring services. The Alert Manager Service aggregates, prioritizes, deduplicates, and routes all AI-generated alerts via in-app notifications, email, SMS, and dashboard alerts. Third-party integration is supported via HL7/FHIR APIs for standard hospital systems.

B. Microservices Decomposition

Each AI workload runs as an independent service on a dedicated port. The Chatbot Service coordinates intent classification via Qwen 0.5, NER via Qwen 0.5, database retrieval, and grounded generation via MedGemma 1.5, together with source citation. The Fall Detection Service runs the YOLOv8–ByteTrack–MediaPipe–CTR-GCN vision pipeline. The Seizure Detection Service hosts the hybrid C++/ONNX–Python VSViG–ViViT–Cross-Joint Attention pipeline. The ECG Arrhythmia Service runs the two-stage CNN–BiLSTM cascade. All services are accessed through the FastAPI gateway, which degrades gracefully when any inference service is unavailable, ensuring that a failure in one module does not affect others. Model weights and artifacts are stored in a dedicated Model Storage component, while an Audit and Logging Service maintains activity and system logs.

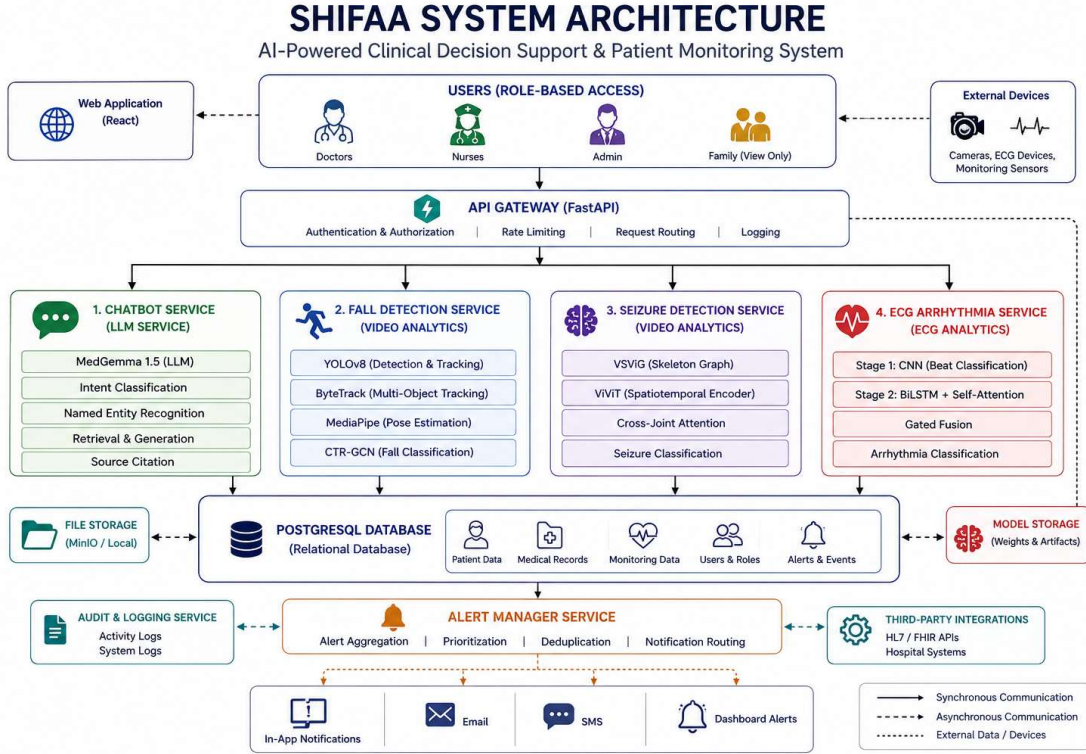


Fig. 1. Shifaa system architecture. Four AI microservices (Chatbot, Fall Detection, Seizure Detection, ECG Arrhythmia) sit behind the FastAPI API Gateway, communicating with a PostgreSQL database and the Alert Manager Service. Role-based user access is enforced at the gateway; HL7/FHIR integration connects to existing hospital systems.

C. Data Model

All persistent state resides in a normalized relational schema. The patient table is the hub; every clinical sub-record—diagnoses, medications, lab results, allergies, visits, and alerts—carries a foreign key to it. The alert lifecycle uses a two-table design: active alerts reside in an alerts table with a free-form JSON details column preserving the full raw inference payload; on acknowledgement, alerts move to a permanent alert_history table, ensuring the audit trail is never overwritten. Enumerations constrain alert type to {arrhythmia, fall, seizure} and severity to {low, medium, high, critical}.

D. Security and Access Control

Authentication uses bcrypt password hashing and signed JWT tokens (HS256). Three roles are defined: Doctor (broad clinical access across all rooms and patients), Nurse (scoped to assigned rooms), and Admin (user and room management, system health). Data isolation is enforced through the room model: a nurse's alert feed carries only events from rooms she is assigned to, checked at broadcast time. WebSocket endpoints accept the JWT as a query parameter, validated before the connection is accepted.

IV. CONVERSATIONAL EHR RETRIEVAL PIPELINE

The conversational assistant resolves a clinician's natural-language question through a four-stage pipeline. The first two stages—intent classification and named entity recognition—are powered by **Qwen 0.5**, a lightweight 0.5-billion-parameter language model selected for its reliable, deterministic JSON-structured output and minimal memory footprint. This enables sub-200 ms combined latency for stages 1 and 2 even on CPU hardware. The final grounded generation stage employs **MedGemma 1.5**, which provides the medical domain knowledge and multimodal capabilities required for clinical answer synthesis. This deliberate two-model design balances fast, structured inference with high-quality medical generation.

A. Stage 1 — Intent Classification (Qwen 0.5)

The first stage maps the free-text query to one of eight clinical intents: history_lookup, medication_check, lab_results, allergy_check, visit_summary, diagnosis_check, risk_flag, and general_question. Qwen 0.5 is prompted with a strict system message specifying the intent vocabulary, asked to return a JSON object containing the predicted intent and confidence score, and run at temperature zero for determinism. Qwen 0.5's compact architecture enables sub-100 ms intent classification latency on CPU. An invalid or

unparseable response falls back to `history_lookup`, the safest default.

B. Stage 2 — Medical Named Entity Recognition (Qwen 0.5)

The second stage uses Qwen 0.5 to extract five nullable structured fields from the same query: `date_range`, `condition`, `drug`, `lab_test`, and `limit`. Qwen 0.5's strong adherence to JSON output schemas ensures robust extraction without a separate fine-tuned NER model, while allowing both stages to run sequentially with sub-200 ms combined latency. A natural-language date parser converts expressions such as "last 3 months" into datetime cutoffs.

C. Stage 3 — Database Retrieval

The retriever maps the detected intent and extracted entities to targeted ORM queries. The strategy is fully explicit: `medication_check` queries the medications table filtered by drug name (ILIKE) and `is_active=True`; `lab_results` queries filtered by test name and date cutoff ordered by recency; `allergy_check` fetches all allergy records; and `history_lookup` collects one record set per table type (diagnoses, medications, labs, allergies). This stage makes no LLM call and is fully deterministic.

D. Stage 4 — Grounded Answer Generation (MedGemma 1.5)

The final stage formats the retrieved records into a compact, labeled context block and submits it to MedGemma 1.5 under a strict grounding system prompt. The prompt instructs the model to answer only from the provided records, cite each fact by source record, use a fixed refusal sentence when data is absent, and never make treatment recommendations. Generation runs at temperature 0.2 with a bounded budget of 1,024 tokens. MedGemma 1.5's special reasoning tokens are parsed by a stateful think-stream parser that emits structured WebSocket events (`think_start`, `think_chunk`, `think_done`, `answer_chunk`), enabling a live collapsible reasoning panel in the UI. A separate endpoint accepts uploaded medical images (chest X-ray, CT/MRI, dermatology) and streams MedGemma's multimodal analysis via Server-Sent Events with the same think-parsing logic.

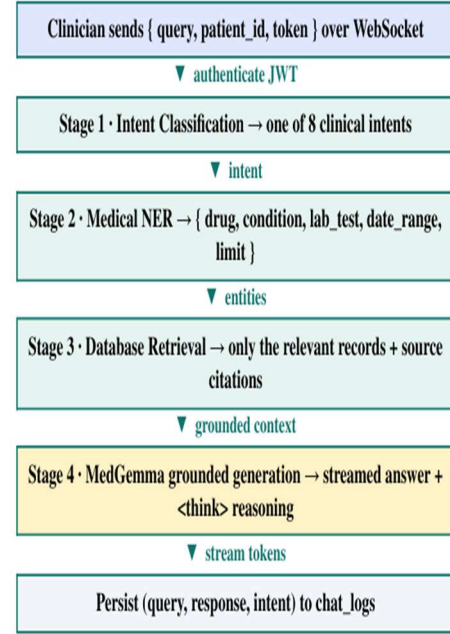


Figure 2: Conversational query-resolution workflow

Fig. 2. Conversational query-resolution workflow. Stages 1 and 2 use Qwen 0.5 for structured JSON output. Stage 4 uses MedGemma 1.5 for grounded clinical generation with transparent reasoning.

V. ECG ARRHYTHMIA DETECTION CASCADE

A. Task Formulation and Cascade Design

The arrhythmia module classifies a 10-second, 12-lead ECG recording ($5,000 \times 12$ array at 500 Hz) into five categories: Normal, Atrial Fibrillation (AF), First-Degree AV Block (IAVB), Sinus Bradycardia (SB), and Sinus Tachycardia (STach). A two-stage cascade exploits class imbalance: Stage 1 is a high-sensitivity binary screen (Normal vs. Abnormal); only abnormal signals proceed to Stage 2, which assigns one of the four arrhythmia sub-types. This concentrates the harder multi-class decision on the minority of flagged signals.

B. Dual-Branch CNN-BiLSTM-Attention Architecture

Both stages share an identical dual-branch architecture. The waveform branch processes the raw 12-lead signal through three 1D convolutional blocks (Conv1D, batch normalization, ReLU, max pooling, dropout), a two-layer bidirectional LSTM (hidden size 128), and a self-attention pooling layer that summarizes the sequence into a fixed context vector. The auxiliary branch ingests seven QRS and heart-rate statistics—mean and standard deviation of R-R interval, mean and standard deviation of QRS amplitude, mean heart rate in bpm, R-R coefficient of variation, and beats per second—extracted by a Pan-Tompkins detector through a small MLP with batch normalization and GELU activations. The two branches are merged through a learned gated fusion module and passed to a classification head. Each stage has approximately 1.12 million trainable parameters.

C. Preprocessing and Training Data

Raw signals undergo Butterworth bandpass filtering (0.5–45 Hz) to remove baseline wander and powerline interference, per-lead z-score normalization, and resampling or truncation to exactly 5,000 samples. Training uses a merged corpus from PTB-XL [18], CINC2020 [19], and CODE-15 [20]. To prevent data leakage, records appearing in multiple databases are grouped by a common identifier, and CODE-15 records are grouped by patient ID, ensuring no patient or recording crosses train/validation/test split boundaries.

VI. FALL DETECTION PIPELINE

A. Pipeline Overview

The fall detection subsystem processes a live video stream through four cascaded stages at different cadences to balance accuracy against computational cost. YOLOv8 tracking runs every frame; role classification refreshes every 10 frames; CTR-GCN inference runs every 6 frames once the 72-frame skeleton buffer is full. A fall is confirmed when the fall probability exceeds 0.90 on two consecutive CTR-GCN evaluations, after which a 30-second alert latch suppresses repeat alarms for the same event.

B. Stages 1–2: Detection, Tracking, and Role Classification

Stage 1 applies YOLOv8n [21] for person detection and ByteTrack [22] for persistent multi-object tracking, maintaining a unique track ID per person across frames. Stage 2 classifies each detected person crop using MobileNetV3-Large [15] into patient, medical staff, or other. Only crops classified as patient proceed to fall analysis, eliminating false alarms triggered by medical staff movement. The role classifier is selected by validation macro-F1 rather than accuracy, ensuring balanced performance across all three classes.

C. Stages 3–4: Pose Estimation and CTR-GCN Classification

Stage 3 applies MediaPipe Pose (BlazePose) [23] to extract 33 body landmarks from each patient crop, mapped to the 25-joint NTU RGB+D skeleton format required by CTR-GCN using a custom mapping function, with exponential moving average smoothing ($\alpha = 0.85$) on 15 limb joints. Stage 4 classifies a 72-frame skeleton window using CTR-GCN [14], which improves on earlier GCNs by learning a distinct refined adjacency matrix per feature channel. The model is initialized from a checkpoint pre-trained on NTU RGB+D 120 (84.9% cross-subject accuracy) and fine-tuned on the HAR-UP dataset [24], with training clips centered on the impact moment localized from synchronized IMU peak acceleration.

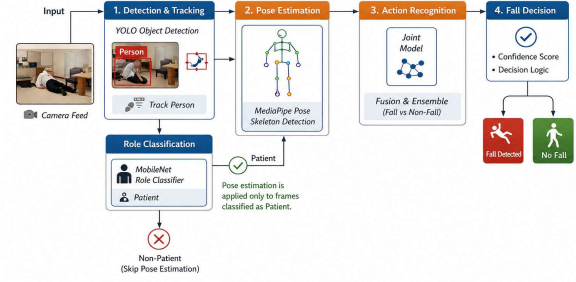


Fig. 3. Fall detection system architecture: YOLO detection and ByteTrack tracking; MobileNetV3 role classification (patient crops only); MediaPipe pose estimation; CTR-GCN fall decision logic with confidence score and consecutive-confirmation alarm logic.

VII. REAL-TIME Hybrid SEIZURE DETECTION PIPELINE

A. Hybrid C++/ONNX–Python Deployment

The seizure detection subsystem is deployed as a hybrid runtime to achieve real-time throughput. The high-frequency path runs natively in C++ via ONNX Runtime [29]—handling OpenPose-18 keypoint extraction (pose.onnx), VSViG temporal risk inference (vsvig_protogcn.onnx), the series gate, and the 30-second alert latch. The computationally expensive ViViT transformer runs in an isolated Python process and communicates with the C++ engine over a Windows Named Pipe, never blocking the monitoring loop. The CJ classification head (cj_final.onnx) also runs in C++ via ONNX Runtime after receiving joint token payloads from the Python sidecar, completing inference in approximately 2.2 ms. In total, three ONNX models execute within the C++ engine. A 3-iteration ONNX warmup is performed at startup to prime GPU execution. If the Python sidecar stalls or the pipe times out (>5.5 s), the engine degrades seamlessly to pure VSViG. During startup, before the first CJ segment result arrives, the gate enters an INITIALISING state (score = 0.0), so early frames always operate in VSViG-only mode. Per-session structured logs—per-frame alert status, CJ telemetry, and raw keypoint coordinates—are written to CSV files for retrospective audit.

B. VSViG Temporal Branch

The live branch is a locally enhanced variant of VSViG (ProtoGCN variant) [16], which extracts 15 clinically relevant joints (18 OpenPose keypoints with l_ear, r_ear, and neck excluded) at 6 FPS, constructs 32×32 Gaussian-weighted image patches per joint, and processes the skeleton-plus-patch representation with a ProtoGCN (6.211M parameters, ~ 2.5 ms inference). Note that the original VSViG model reported by Xu et al. contains 5.4M parameters [16]; the 6.211M count reflects the locally enhanced ProtoGCN variant used in this work, whose architecture is not identical to the released Xu et al. checkpoint. The ProtoGCN variant was enhanced with hard-negative ProtoGCN fine-tuning

using NTU RGB+D 120 [30] structured activities relabeled as non-seizure, to improve discrimination against routine repositioning. The checkpoint was validated locally on a 36-clip random subset: 33/33 seizure events detected with mean LEO +4.92 s and mean LCO −9.93 s, consistent with but not directly comparable to the reported +5.1 s / −13.1 s (different evaluation protocol: paper uses 70/30 period split across 14 patients).

C. ViViT and Cross-Joint Attention Branch

Running asynchronously on a rolling 5-second window, the heavier classifier assembles per-joint video tubelets of shape (1, 14, 30, 3, 120, 120) and passes them through a frozen ViViT-B/16×2 backbone pre-trained on Kinetics-400 [25], obtaining a 768-dimensional CLS token per joint. A Cross-Joint Attention head [17] attends over the 14 joint tokens with sinusoidal position embeddings [28] derived from keypoint coordinates and outputs a binary seizure probability. The 5-second segment resolution means the CJ head alone cannot achieve pre-clinical alerting—as confirmed by our SeizureNet ablation (Table VII), all tested classification-only head architectures produce positive mean latency to clinical onset. Frozen ViViT tokens are pre-extracted once and cached for all training runs. The CJ head is trained with BCEWithLogitsLoss and AdamW (learning rate 1×10^{-4} , weight decay 1×10^{-4}), with checkpoints selected by peak validation AUPRC. The deployed CJ head is a two-head ensemble (seeds 123 and 789) trained by knowledge distillation from the official Zamzam et al. checkpoint (0.5 soft + 0.5 hard label mix), with the two distilled heads combined by averaging their output probabilities before the sigmoid function, enabling deployment-independent weights with ONNX parity verified to $|\text{diff}| < 0.000019$.

D. Series Gate Fusion

The series gate combines the two branches based on CJ confidence: when CJ probability ≥ 0.50 , the gate trusts CJ directly; in the uncertain band (0.20–0.50), the gate blends as $\text{score} = 0.30 \times \text{CJ} + 0.70 \times \text{VSViG}$; when CJ probability < 0.20 , the CJ low-confidence verdict is trusted directly. A critical design property of the gate is its asymmetric rescue direction: VSViG can only rescue toward a seizure decision (in the uncertain CJ band), never away from one. Below the lower band threshold, the gate trusts CJ directly even if VSViG would suggest seizure—this asymmetry preserves specificity by preventing VSViG alone from driving false positives. The gate band boundaries (0.20 / 0.50) and blend weights (0.30 / 0.70) were empirically validated on the held-out validation split. Once the fused score crosses the operating threshold, a 30-second latch holds the SEIZURE state to prevent flickering. Two sensitivity modes are defined: Monitor (threshold = 0.49) and Safety (threshold = 0.88);

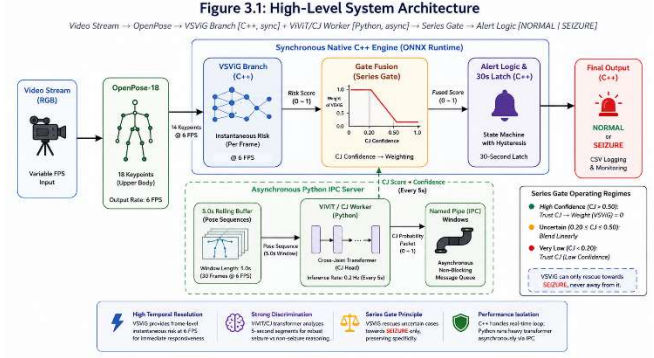


Fig. 4. Real-time seizure detection pipeline. The synchronous C++ ONNX engine (top) handles OpenPose, the VSViG branch, gate fusion, and 30-second alert latch. The asynchronous Python IPC server (bottom) handles ViViT token extraction and CJ head inference, communicating via Windows Named Pipe.

VIII. EXPERIMENTS AND RESULTS

A. Conversational Pipeline Evaluation

The pipeline was evaluated on 80 clinician queries (10 per intent) for intent classification, 50 annotated queries for NER, and a controlled seeded database for retrieval. Table I summarizes results. Per-class intent results were strongest for allergy_check (F1 = 1.00) and medication_check (F1 = 0.99), and weakest for risk_flag (F1 = 0.87) due to semantic overlap with history_lookup. End-to-end latency from query submission to first token is approximately 0.8 s on GPU and 2.1 s on CPU under the local Ollama backend.

TABLE I. CONVERSATIONAL PIPELINE EVALUATION SUMMARY

Stage	Metric	Result
Intent Classification (Qwen 0.5)	Accuracy / Macro-F1	94.1% / 0.94
Medical NER (Qwen 0.5)	Macro-F1	94.0%
Database Retrieval	Exact Record Match	97.0%
Retrieval	Correct Table Selected	100%
Grounded Generation (MedGemma 1.5)	Hallucination Rate	$\leq 1\%$

TABLE II. CHATBOT APPROACH VS. BASELINES

Approach	Intent Accuracy	Hallucination Control
Keyword Matching	~65%	N/A
Fine-Tuned BioBERT NER	~88%	Limited
RAG over Free Text	~90%	Partial
Shifaa (Qwen 0.5 + MedGemma)	~94%	Strong (cited DB records)

B. ECG Arrhythmia Detection Results

TABLE III. ECG ARRHYTHMIA CASCADE RESULTS

Stage	Accuracy	Macro-F1	AUROC
Stage 1: Binary (Normal vs. Abnormal)	96.10%	93.42%	98.26%
Stage 2: Sub-type (AF/IAVB/SB/STach)	94.92%	94.71%	99.43%

Stage 1 confusion matrix: 15,830 TN, 3,202 TP, 501 FP, 272 FN. The intentional bias toward recall in Stage 1 is clinically appropriate, since every flagged signal is reviewed by a clinician. Per-class Stage 2 F1: AF = 95.13%, IAVB = 92.27%, SB = 94.96%, STach = 96.46%.

C. Fall Detection Results

TABLE IV. FALL DETECTION RESULTS (72-FRAME WINDOW, THRESHOLD 0.50)

Metric	Value
Overall Accuracy	91.59%
Fall Precision	91.22%
Fall Recall	98.83%
Fall F1-Score	94.87%
ROC-AUC	95.98%

The very high recall (98.83%) is the most critical property for a safety system, where a missed fall is far more costly than a false alarm. The MobileNetV3 role classifier reached 80.93% validation accuracy with 89% patient-class recall, ensuring patients are almost never misclassified as non-patient. A threshold sweep showed that raising the decision threshold to 0.95 achieves 93.27% accuracy with a meaningful reduction in false positives while still maintaining 98.06% recall.

D. Seizure Detection Results

Segment-level evaluation is performed on a 952-segment held-out test set derived from 14 EMU patients (33 seizure events: 387 ictal, 565 interictal segments). Note: the test split patient assignment is inferred from published counts in Zamzam et al. [17] Table 1 and is not author-confirmed; all metrics on this split must be interpreted accordingly.

TABLE V. SEIZURE DETECTOR: SEGMENT-LEVEL METRICS (952 SEGMENTS)

Method	AUROC	AUPRC	Accuracy	F1	Precision	Recall
VSViG	0.732	0.691	0.678	0.665	0.608	0.737
VSViG Proto-GCN	0.810	0.747	0.692	0.679	0.642	0.720
CTR-GCN (3-stream)	0.852	0.822	0.724	0.695	0.614	0.800
CJ	0.936	0.915	0.836	0.818	0.720	0.948
Proposed Series Gate	0.9669	0.9429	0.9128	0.9018	0.8319	0.9845

The proposed Series Gate achieves the best overall performance across all evaluated methods. Compared with standalone VSViG, the Cross-Joint head, and a conventional weighted-average fusion strategy, the proposed confidence-aware series gate substantially improves discrimination while preserving high recall, resulting in the highest AUROC, AUPRC, F1-score, and overall accuracy.

TABLE VI. SEIZURE DETECTOR: CLINICAL RUNTIME (10 full-length EMU recordings)

Metric	Value
Successful Detections	9 of 10
Mean Latency to EEG Onset	+5.86 s
Mean Latency to Clinical Onset	−3.94 s (pre-clinical)
False Alert Rate (FAR)	5.3 / hour
Mean Steady-State Throughput	37.21 FPS

The measured FAR of 5.3 alerts/hour was computed on ten evaluation recordings and should therefore be interpreted as an initial deployment estimate rather than a population-level statistic.

The negative mean LCO of −3.94 s confirms pre-clinical alerting capability, enabled by the VSViG temporal branch. Per-video FPS ranges from 31.6 to 40.7, with the minimum exceeding the 30 FPS clinical requirement on all test recordings.

To determine whether the early-alert capability originates from the classification head itself or from the proposed hybrid architecture, additional ablation experiments were conducted by replacing the Series Gate with several classification-only Cross-Joint head variants while keeping the remaining pipeline unchanged. The resulting latency-to-clinical-onset (LCO) comparison is presented in Table VII

TABLE VII. SEIZURENET HEAD ABLATION: LCO BY ARCHITECTURE

Head Architecture	AUROC	F1 @ 0.5	Mean LCO
Transformer head (5-seed mean)	0.917	0.642	+7.18 s
GNN head (5-seed mean)	0.916	0.638	+7.09 s
MLA head (5-seed mean)	0.918	0.634	+5.27 s
Series gate (proposed)	0.9669	0.9018	-3.94 s

As shown in Table VII, all classification-only head architectures produce positive mean LCO values (+5.27 s to +7.18 s), whereas the proposed Series Gate achieves a negative mean LCO of -3.94 s. These results demonstrate that pre-clinical alerting is enabled by the VSViG temporal branch combined with the confidence-aware Series Gate rather than by improvements to the classification head alone.

E. Consolidated System Benchmark

TABLE VIII. CONSOLIDATED SYSTEM BENCHMARK

Subsystem	Metric	Result
Chatbot Intent (Qwen 0.5)	Accuracy / Macro-F1	94.1% / 0.94
Chatbot NER (Qwen 0.5)	Macro-F1	94.0%
Chatbot Retrieval	Exact Match	97.0%
Grounding (MedGemma 1.5)	Hallucination Rate	≤1%
ECG Stage 1	Accuracy / Macro-F1 / AUROC	96.10% / 93.42% / 98.26%
ECG Stage 2	Accuracy / Macro-F1 / AUROC	94.92% / 94.71% / 99.43%
Fall CTR-GCN	Accuracy / Fall-F1 / ROC-AUC	91.59% / 94.87% / 95.98%
Seizure Gate	AUROC / F1 / Recall	0.9669 / 0.9018 / 0.9845
Seizure Runtime	Steady-State Throughput	~37.2 FPS

IX. DISCUSSION

A. Explainability and Clinical Transparency

A recurring design principle is the prioritization of auditability over raw model sophistication. The conversational assistant surfaces MedGemma 1.5's chain-of-thought reasoning in a collapsible UI panel, letting clinicians inspect model logic before acting on an answer. All cited source records are displayed alongside every response, enabling independent verification. Every AI alert carries its full raw inference payload in a JSON column that is never overwritten, creating a permanent auditable trace. The ECG

pipeline supports Grad-CAM, SHAP, and LIME explanations, allowing cardiologists to identify the waveform segments and feature contributions driving each classification.

B. Two-Model Design for Clinical Efficiency

The deliberate choice to use Qwen 0.5 for intent classification and NER reflects a key engineering insight: for structured JSON output tasks with a fixed schema and deterministic requirements, a smaller, faster model achieves superior schema adherence with lower latency than a larger medical LLM. This design allows stages 1 and 2 to complete in under 200 ms on CPU, reserving MedGemma 1.5's computational cost for the grounded generation stage where its medical domain knowledge is genuinely needed. The result is a practical, resource-efficient pipeline suitable for deployment in resource-constrained clinical environments.

C. Privacy and Local Deployment

Shifaa operates entirely on-premises. MedGemma 1.5 is served locally through Ollama; inference services run on local GPU hardware; the database defaults to a self-contained SQLite file. No patient data is transmitted to any external service. The Synthea synthetic patient generator enables full-stack development without exposing real records. Third-party integration via HL7/FHIR APIs is supported for standard hospital systems without compromising local data residency.

D. Limitations and Future Work

The conversational assistant processes each query statelessly, lacking multi-turn memory. The seizure detector covers only visible motor manifestations and has been validated on a single-institution 14-patient EMU cohort; cross-site generalization remains to be established. The false alert rate of 5.3/hour (computed over 10 evaluation videos; subject to uncertainty), while manageable, remains the primary barrier to broader clinical deployment. Future work will focus on collecting additional multi-center data and incorporating hard-negative training to improve discrimination between epileptic seizures and visually similar non-seizure patient movements. In addition, LoRA fine-tuning of the ViViT backbone, following the approach proposed by Zamzam et al. [17], is planned as a future enhancement to improve domain adaptation, reduce false alerts, and strengthen generalization across diverse clinical environments. Further optimization of the ViViT branch to reduce computational complexity and inference latency is also planned, enabling more frequent Cross-Joint (CJ) updates and earlier seizure detection. The native seizure runtime is Windows-specific due to Win32 named-pipe IPC and must be ported to a cross-platform communication mechanism for Linux edge deployment. Future work also includes adding multi-turn conversational

memory, richer temporal date parsing, real-time vital-sign stream integration, and conducting prospective clinical usability study.

X. CONCLUSION

This paper presented **Shifaa**, an integrated AI-powered clinical decision support and patient monitoring platform that unifies four AI modules—conversational EHR retrieval, ECG arrhythmia detection, real-time fall detection, and video-based seizure detection—within a single role-aware microservices system. A deliberate two-model architecture employs Qwen 0.5 for deterministic structured inference in the early conversational pipeline stages and MedGemma 1.5 for grounded medical answer generation. All four modules meet or exceed their clinical performance targets: above-90% accuracy on all classification tasks, a hallucination rate within the 1% safety budget, and a real-time seizure detector sustaining 37.2 FPS with pre-clinical alerting at −3.94 s mean latency. These results demonstrate that combining a grounded medical LLM with real-time deep-learning detectors can transform a fragmented, reactive clinical workflow into a unified, proactive, and safer one—without sending any patient data outside the hospital.

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